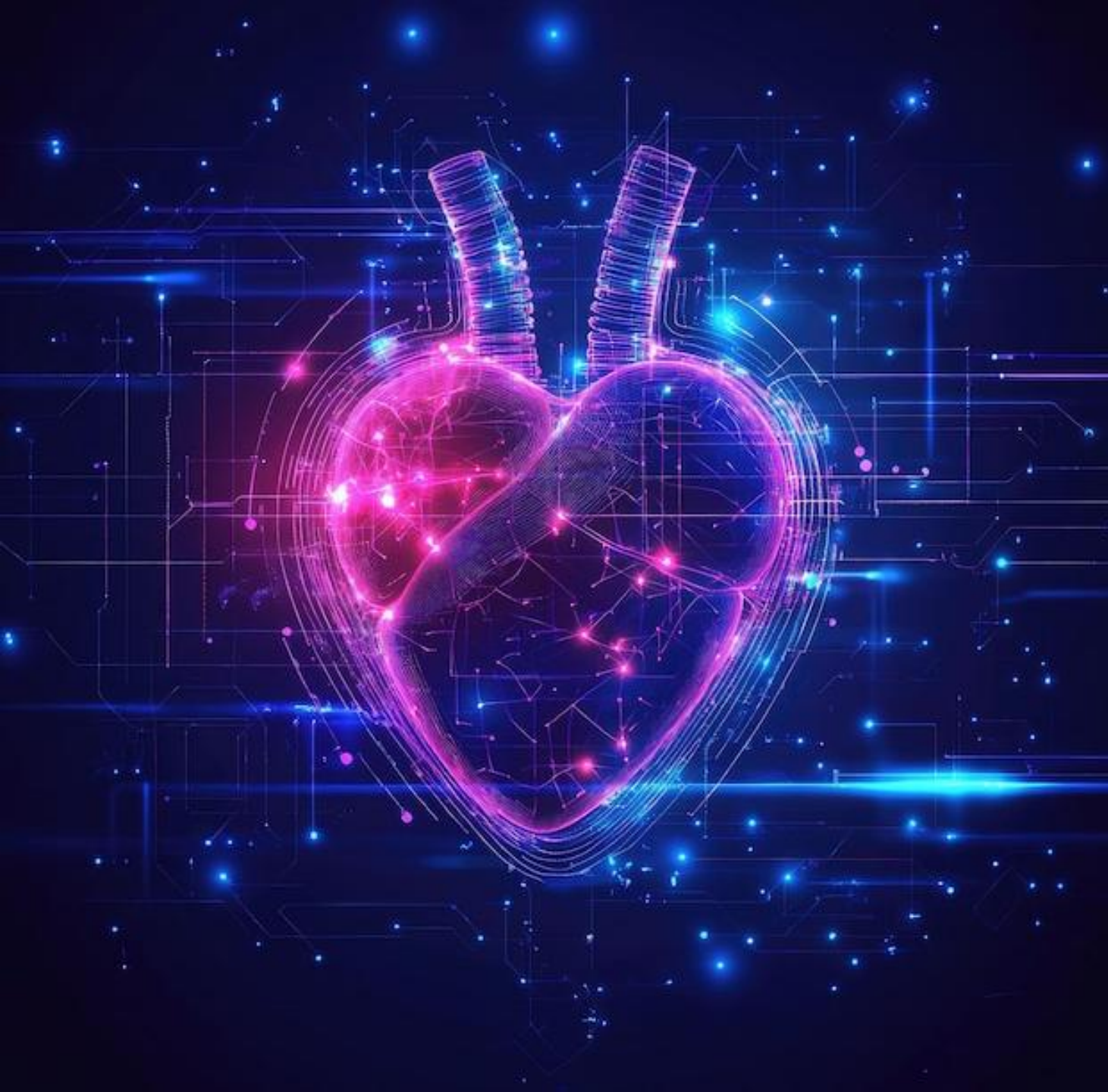




Predicting Heart Disease Risk Using Machine Learning Techniques on Clinical Patient Data

By Owais Jawaid

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Why This Project Matters

- Heart disease = *#1 global cause of death* (~32% of deaths)
- Early detection dramatically improves outcomes
- Machine learning can identify hidden patterns
- Goal: Build a predictive ML model to predict heart disease using routine clinical data



Project Objectives

- Analyze clinical heart disease data
- Clean, preprocess, and engineer useful features
- Train multiple machine learning models
- Compare performance using accuracy & ROC-AUC
- Identify key clinical predictors
- Choose and save the best overall model



Dataset Overview

- Source: Kaggle (heart.csv)
- Initial records: 1025
- After removing duplicates: 302
- No missing values
- Target:
 - 1 = Heart Disease
 - 0 = No Heart Disease
- Distribution:
 - 164 disease
 - 138 no disease

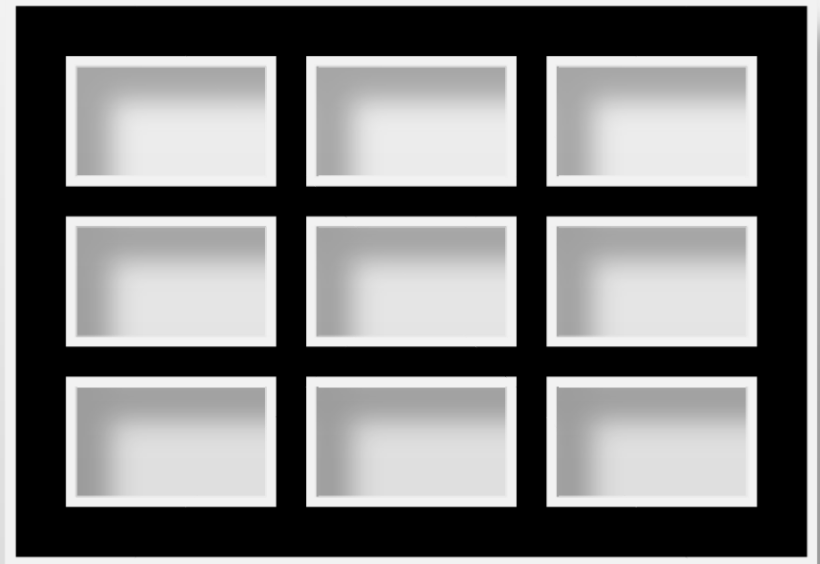


Feature Summary

Key clinical features:

- **cp** — Chest pain type
- **thalach** — Max heart rate
- **oldpeak** — ST depression
- **exang** — Exercise-induced angina
- **ca** — Major vessels seen in fluoroscopy

Other features: age, sex, cholesterol, resting BP, thal, slope, etc.



- **age:** Age of the patient (in years)
- **sex:** Gender (0 = female, 1 = male)
- **cp:** Chest pain type (0 – 3)
 - 0 = typical angina
 - 1 = atypical angina
 - 2 = non-anginal pain
 - 3 = asymptomatic
- **trestbps:** Resting blood pressure
- **chol:** Serum cholesterol (mg/dl)
- **fbs:** Fasting blood sugar > 120 mg/dl (1 = true, 0 = false)
- **restecg:** Resting electrocardiographic results:
 - 0 = normal
 - 1 = ST-T wave abnormality
 - 2 = left ventricular hypertrophy

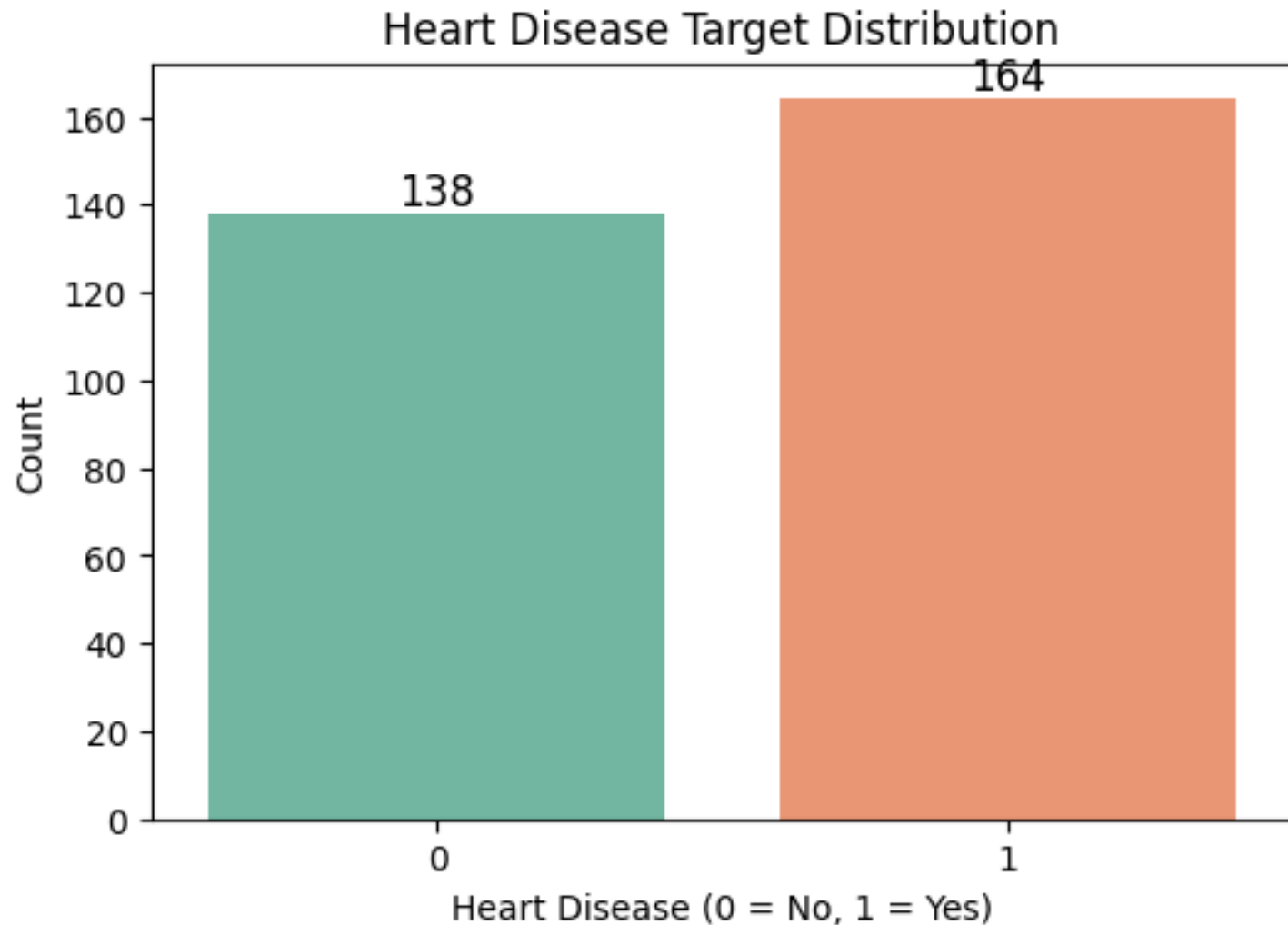
- **thalach:** Maximum heart rate achieved
- **exang:** Exercise-induced angina (1 = yes, 0 = no)
- **oldpeak:** ST depression induced by exercise
- **slope:** Slope of the peak exercise ST segment:
 - 0 = upsloping
 - 1 = flat
 - 2 = downsloping
- **ca:** Number of major vessels (0 – 3) colored by fluoroscopy
- **thal:** Thalassemia (blood disorder type):
 - 1 = Normal
 - 2 = Fixed defect
 - 3 = Reversible defect
- **target:** Target variable
 - 1 = patient has heart disease
 - 0 = patient does not have heart disease

Preprocessing Steps



- Removed duplicates
- Encoded categorical variables
 - cp, thal, slope, sex
- Standardized numeric values
 - chol, trestbps, thalach, oldpeak
- Created **age groups**
 - Young, Middle-Aged, Senior, Elderly
- One-hot encoded categories
- Train-test split →
 - Train (80%): **241** records
 - Test (20%): **61** records

Data Loading and Cleaning Visuals



```
Dataset loaded successfully.  
Initial shape: (1025, 14)  
After removing duplicates: (302, 14)
```

Missing values:

age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalach	0
exang	0
oldpeak	0
slope	0
ca	0
thal	0
target	0

dtype: int64

Exploratory Data Analysis (EDA)

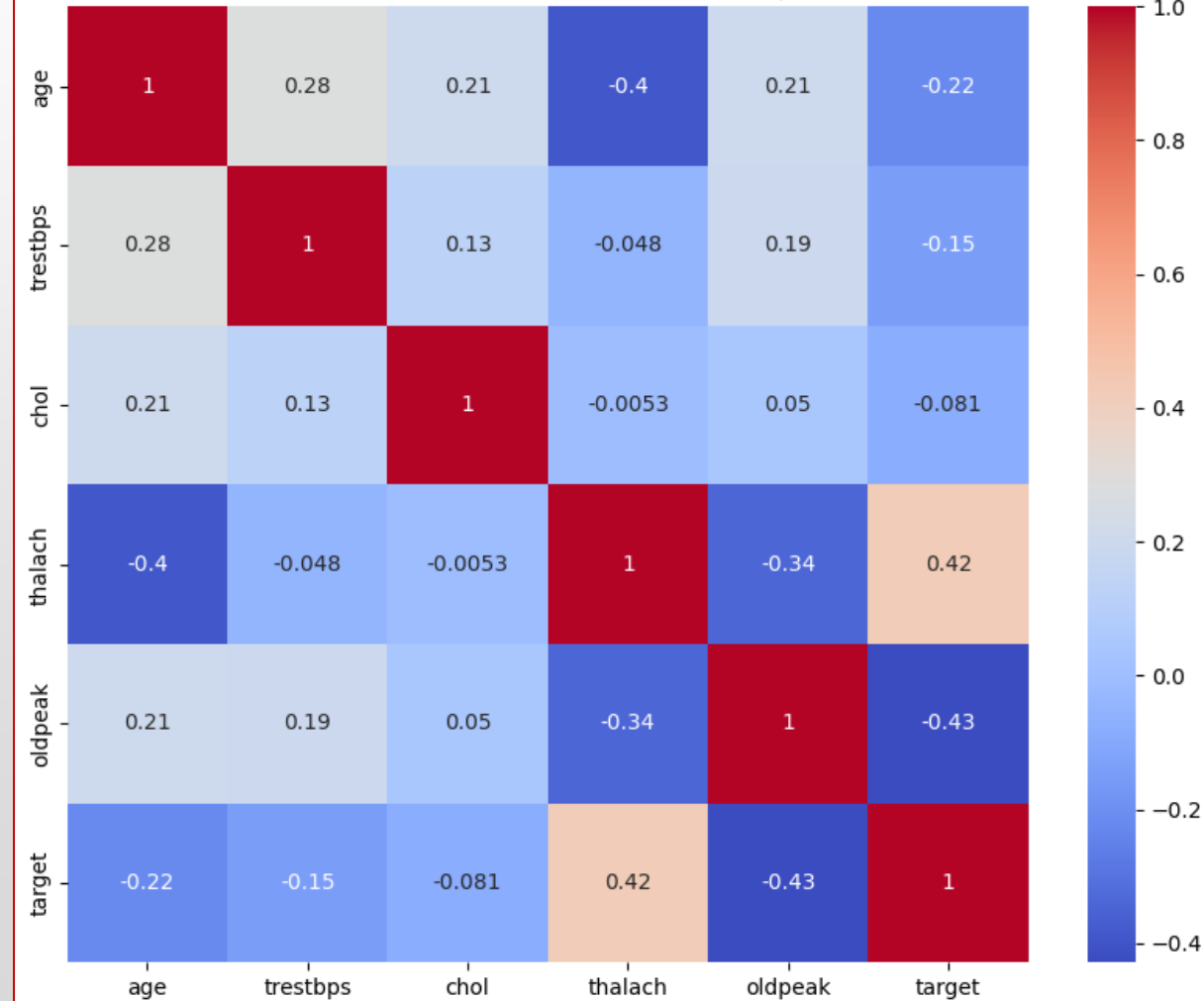
Analyses performed:

- Target distribution
- Feature distributions
- Correlation heatmap
- Chest pain vs disease
- Heart rate vs disease

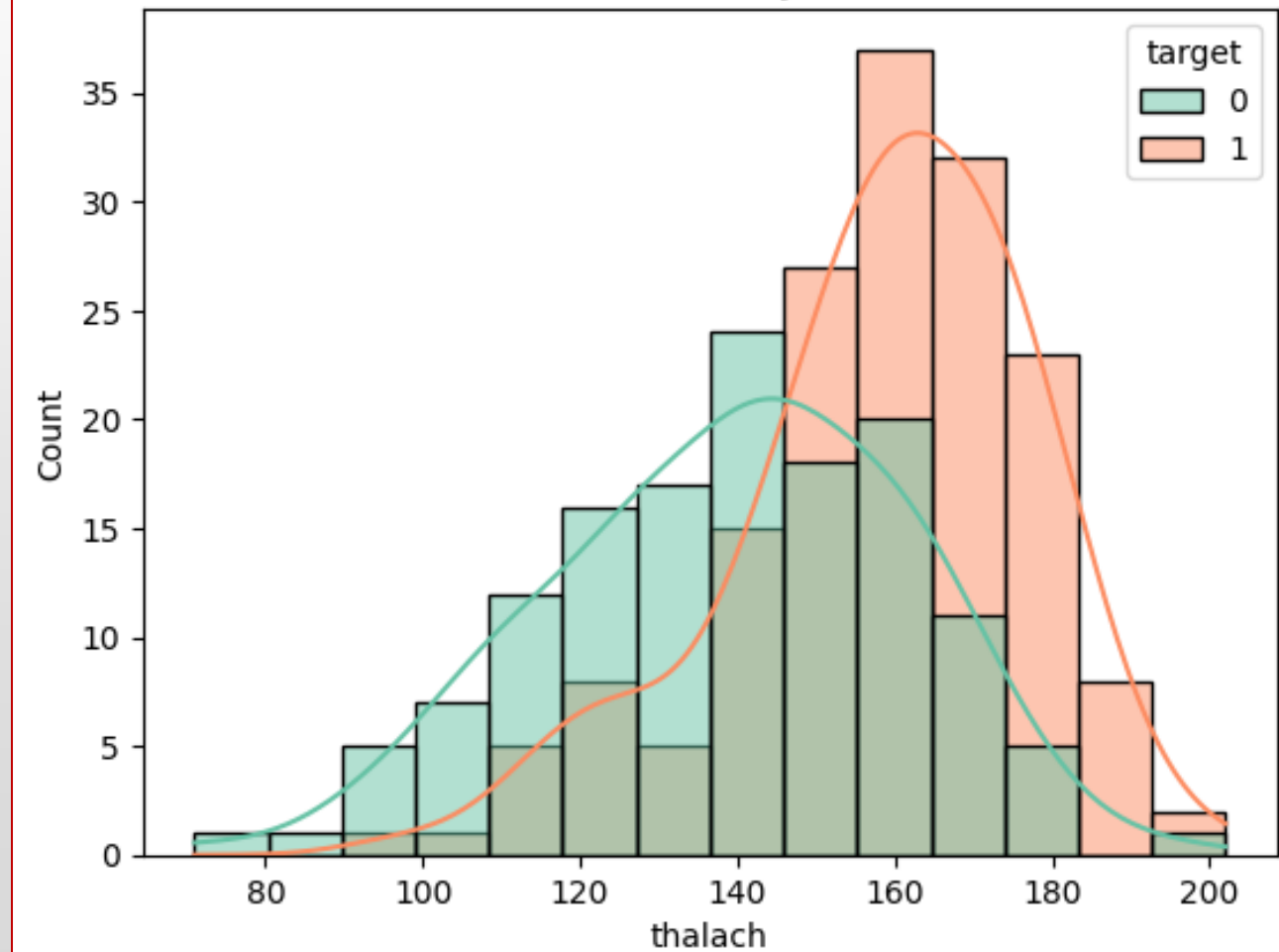


Exploratory Data Analysis (EDA) Visuals

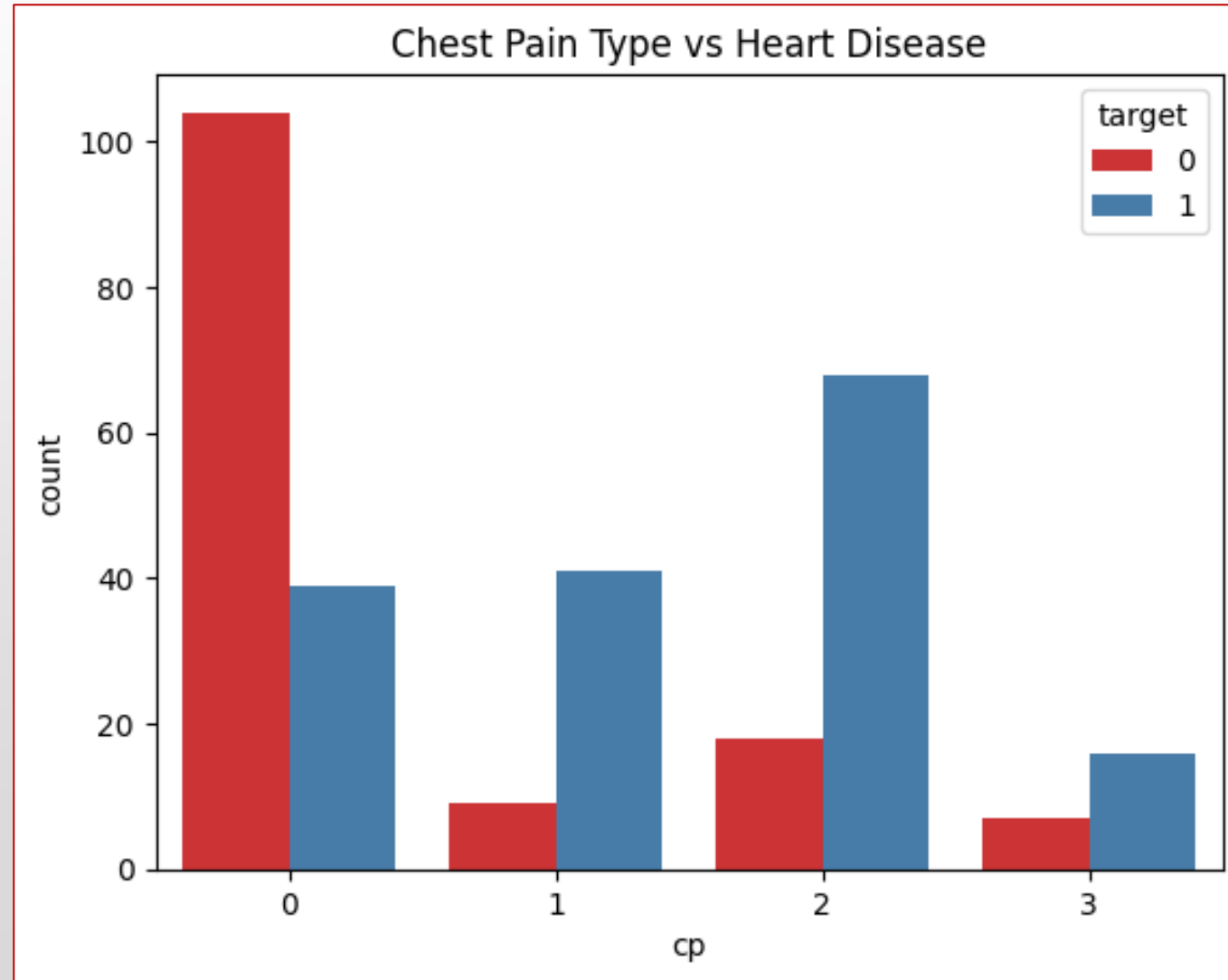
Numeric Feature Correlation Heatmap



Maximum Heart Rate by Heart Disease



Exploratory Data Analysis (EDA) **Visuals** continued...



Key EDA Findings

- Higher chest pain types correlate with heart disease
- Lower max heart rate = higher risk
- Higher oldpeak (ST depression) values appear in disease cases
- Some features show moderate correlation, none dominate



Models Tested

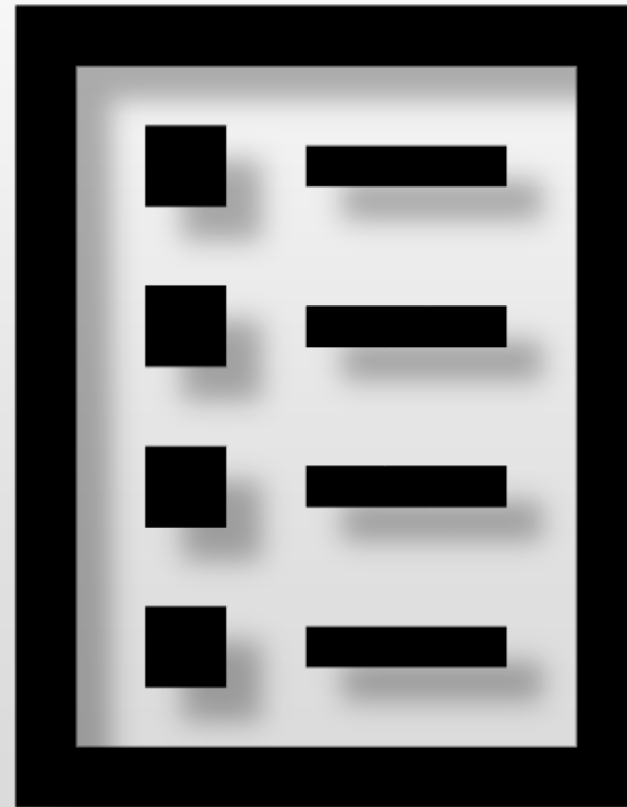
Baseline:

- Logistic Regression
- Naive Bayes

Advanced Models:

- Random Forest
- XGBoost
- Support Vector Machine (SVM)

Hyperparameter tuning done with
GridSearchCV.

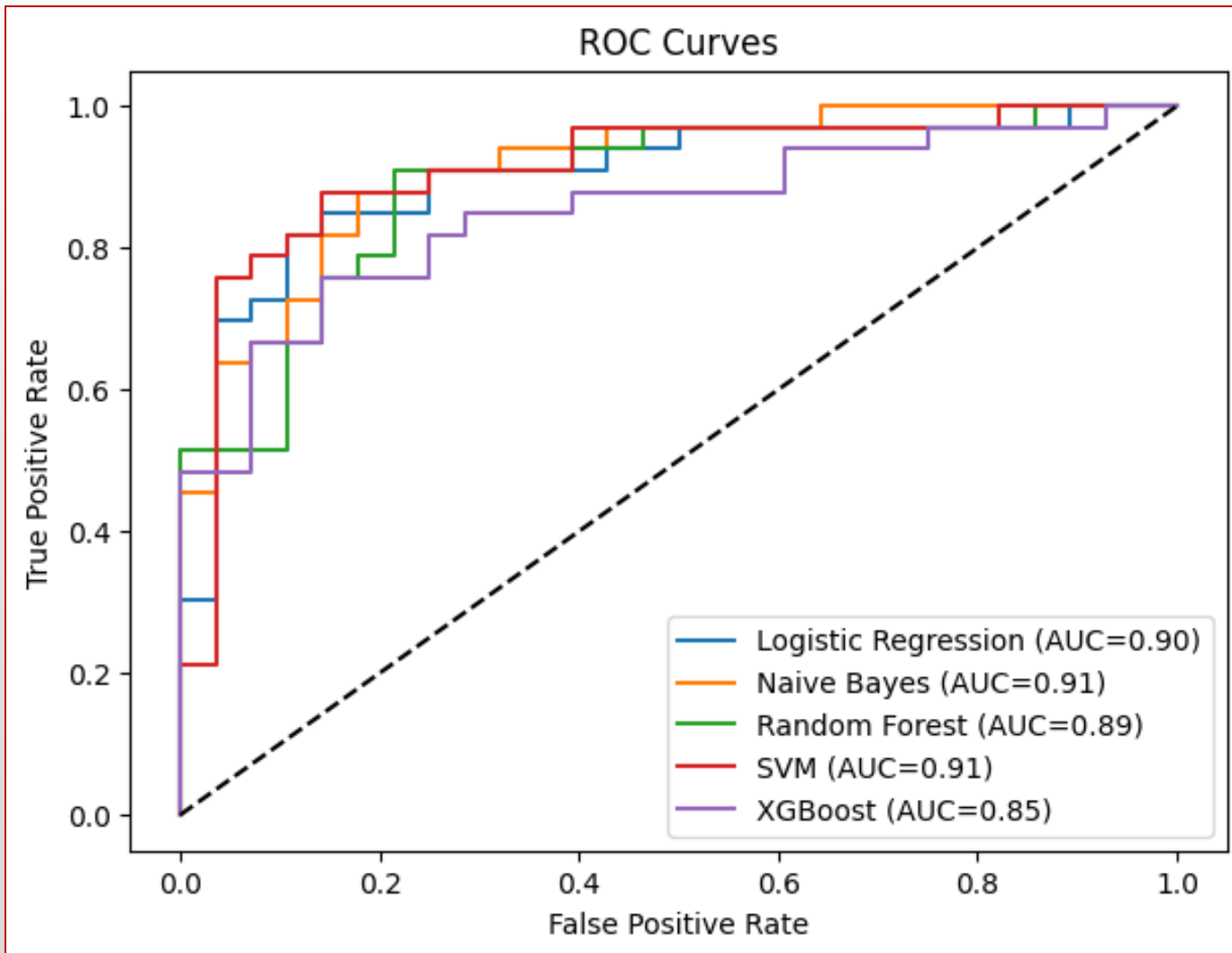


Model Performance Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.852	0.875	0.848	0.862	0.899
Naive Bayes	0.820	0.844	0.818	0.831	0.911
Random Forest	0.787	0.833	0.758	0.794	0.887
XGBoost	0.787	0.833	0.758	0.794	0.851
SVM	0.869	0.879	0.879	0.879	0.910

SVM achieved the highest accuracy and balanced performance across all evaluation metrics.

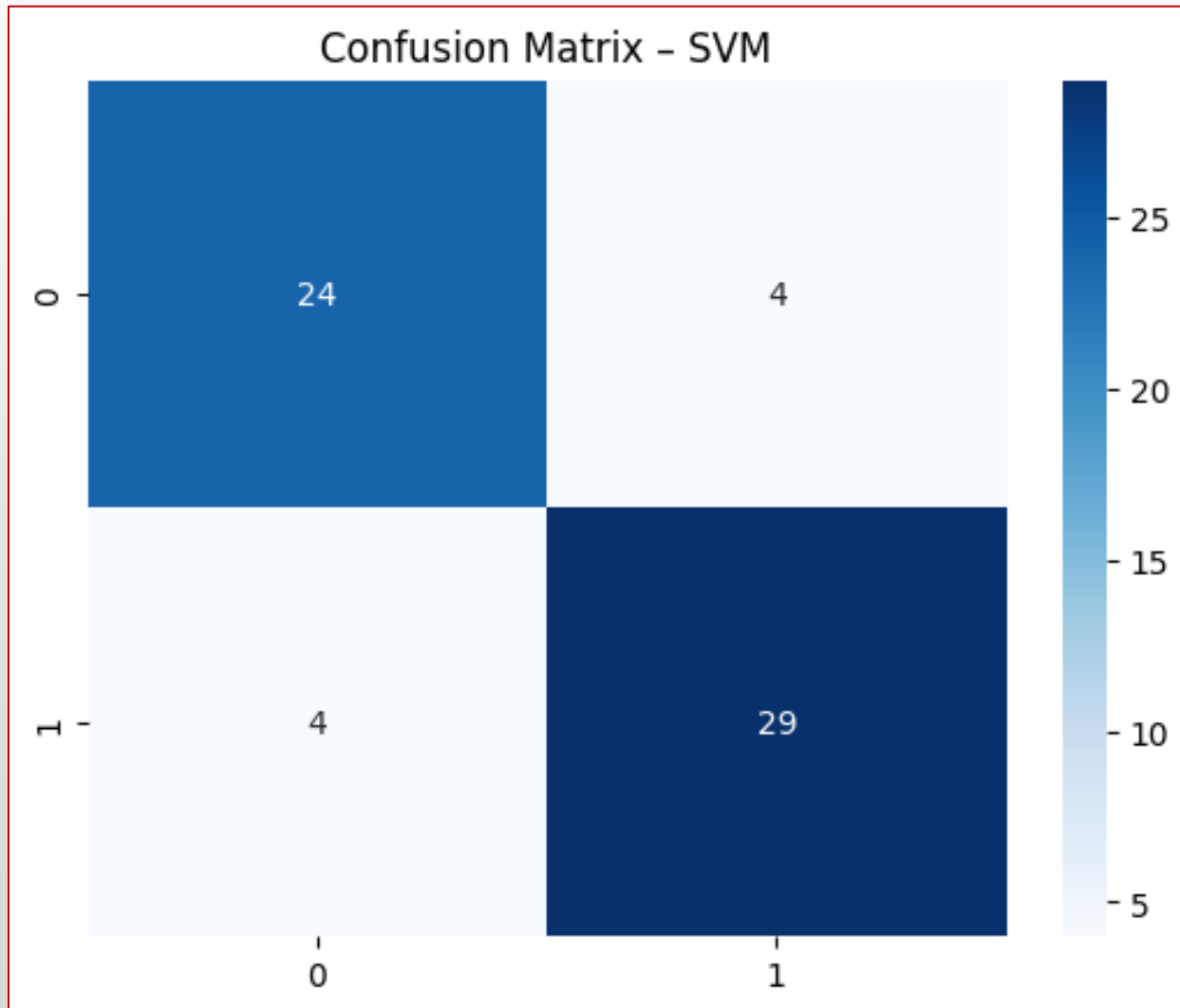
ROC Curves



Highlights:

- SVM and Naive Bayes $\approx$ **0.91 AUC**
- Strong separation between classes

Confusion Matrix (SVM)

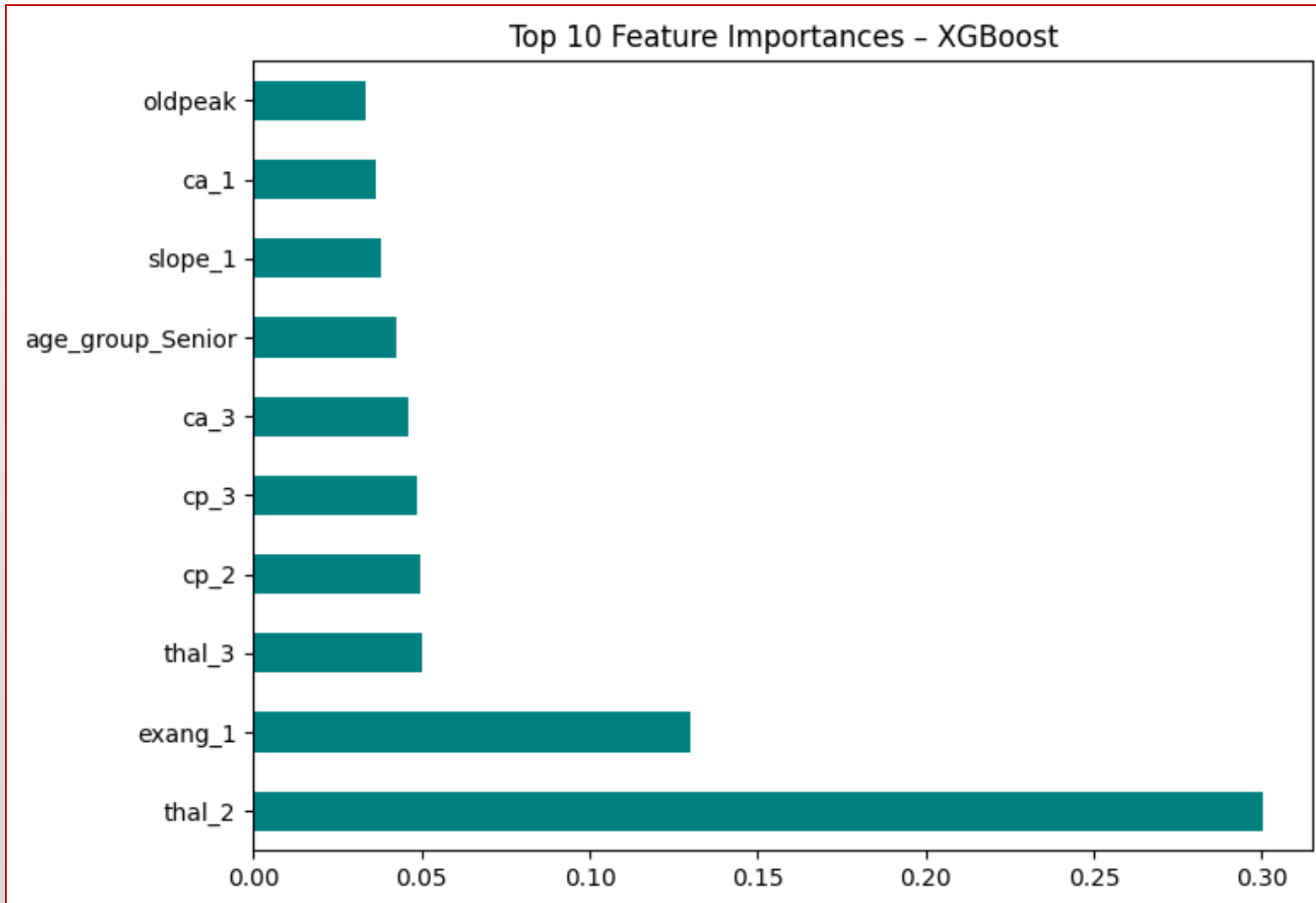


Metrics:

- True negatives: 24
- True positives: 29
- Accuracy: **87%**

SVM misclassified very few patients, proving its reliability.

Important Features



Top predictors:

- **Chest pain type (cp)**
- **Max heart rate (thalach)**
- **ST depression (oldpeak)**
- **Exercise-induced angina (exang)**
- **Number of vessels (ca)**

These factors play the biggest role in predicting heart disease.

Final Model Choice

Chosen Model: Support Vector Machine (SVM)

Why?

- Highest accuracy
- High precision and recall
- Generalizes well
- Works well with standardized clinical data

SVM proved to be the strongest and most stable model overall.



Conclusion

- ML can effectively detect heart disease risk
- SVM proved most reliable
- Key features relate to chest pain, heart rate, and ST depression
- Potential use in clinical decision support

This project shows ML's potential to support early heart disease screening.

Future Enhancements

- Deploy model via a web or mobile app
- Add new clinical features (ECG scans, imaging)
- Expand dataset
- Test on real patient data

With more data and deployment, this model could support real clinical workflows.

References

- Lapp, D. (2019, June 6). Heart disease dataset. Kaggle. <https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset/data#>
- Rawat, S. (2025, January 23). *Heart disease prediction*. Towards Data Science. <https://towardsdatascience.com/heart-disease-prediction-73468d630cfc/>
- World Health Organization. (2025, July 31). Cardiovascular diseases (CVDs). World Health Organization. [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds))

THANK YOU

FOR YOUR ATTENTION

